Md Zesun Ahmed Mia

Device-Circuit Co-Design | Spintronics & Neuromorphic Systems

Research engineer focused on the intersection of novel devices and circuit design, experienced in creating robust models and prototypes for emerging spintronic and neuromorphic systems.

Education

08/2022–05/2027 (Expected) **Ph.D. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, *CGPA: 4.00/4.00*
Advisor: Dr. Abhronil Sengupta

08/2022–08/2024 **M.S. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, *CGPA: 4.00/4.00* | Thesis: Neuromorphic Computing for Lifelong Learning

02/2015–04/2019 **B.Sc. in Electrical and Electronic Engineering**, *Bangladesh University of Engineering & Technology (BUET)*, Dhaka, Bangladesh, *CGPA: 3.81/4.00*

Industrial Experience

05/2025–07/2025 **Graduate Technical Intern**, *Intel Corporation*, Hillsboro, OR

- Optimized thin film deposition processes affecting microprocessor circuit performance and electrical characteristics for advanced logic devices, bridging device fabrication and circuit design.
- Designed DOE to control critical dimensions (CD) impacting yield and electrical parameters in next-generation processors.
- Applied statistical modeling to predict process impact on circuit performance metrics and device reliability, supporting integrated device-circuit design decisions.

09/2020–07/2021 **R&D Engineer**, *SEMWAVES Ltd.*, London, UK, part-time

- Delivered a 50 kW hybrid renewable system (solar + smallscale hydro) for an offgrid Bangladeshi site, supporting reliable community power.
- Owned technical leadership and coordination (vendors, field teams, stakeholders); directed device selection, integration, QA/safety procedures, and optimization against load profiles and uptime targets.

Technical Skills

Device & Circuit	Spintronic/neuromorphic hardware, SNNs, advanced memory, SRAM/DRAM device-circuit co-design, PIM, near-memory computing, memory-centric computing, memory controller design, process integration, TCAD, compact modeling, parameter extraction, Verilog-A, CMOS/FinFET circuit design, VLSI/IC design, ASIC design flow, Digital logic design, Systolic Arrays, Testbench development
DTCO & STCO	Design-technology and system-technology co-optimization, cross-layer PPA, process-aware design, system-architecture codesign
EDA/Sim.	TCAD, Cadence Virtuoso, Spectre, HSPICE, COMSOL, ModelSim, Synopsys (Design Compiler, PrimeTime, VCS), FPGA prototyping, CIM (Compute-In-Memory) Systems
Programming	Python, CUDA, MATLAB, Verilog, Shell, C/C++, Bash, Linux/Unix
ML/AI Tools	PyTorch, TensorFlow, TensorRT, Data Visualization (Matplotlib, Seaborn, Plotly), Pandas, NumPy, JMP, Jupyter, LaTeX, AI accelerator development, HPC applications, parallel computing architectures
AI Dev.	Generative and agentic AI tools (Cursor, Copilot, etc.) for research, design, and code development
Hardware & Fabrication	PCB design, Oscilloscope, Signal Generator, LabVIEW, Lithography, Etching, Deposition (CVD, HDP-CVD, PVD), E-beam
Characterization & Reliability	AFM, SEM, Probe Station, Electrical Testing (DC/AC characterization, IV/CV measurements), TEM, endurance, retention, failure analysis, Metrology (XRR, XRD, ellipsometry, profilometry)
Collaboration	Git, Microsoft Office, Google Workspace

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Academic Research and Teaching Experience

08/2022– Present **Graduate Research Assistant**, *Penn State*, University Park, PA

- Bridged device physics and circuit design by creating compact Verilog-A models of FeFETs from characterization data for use in HSPICE simulations.
- Led a device-circuit co-design effort by integrating fabricated spintronic devices into a system-level simulation framework using hardware-aware ML training. Implemented hardware-software co-design methodologies for PIM systems, optimizing data flow and memory-centric computing architectures.
- Utilized TCAD device simulations (Sentaurus/Silvaco) to guide the design of FETs, ensuring their electrical characteristics were optimized for target circuit applications.

08/2024– 05/2025 **Graduate Teaching Assistant**, *Penn State*, State College, PA

- Taught Cadence Virtuoso (schematic/layout), PDK usage, DRC/LVS, and analog/digital design flows; created hands-on lab modules and guided tool/debug workflows.
- Supervised Capstone projects for 90+ undergraduate students across communications, electronics, and firmware: supported Raspberry Pi/Arduino development, software-defined radio experiments, PCB design, and system integration end-to-end.

02/2021– 08/2022 **Lecturer**, *University of Liberal Arts Bangladesh (ULAB)*, Dhaka, Bangladesh

- Taught undergraduate courses: Solid State Devices, Digital Circuit Design, Semiconductor Device Physics, Power Electronics.
- Developed lab modules and supervised projects on semiconductor devices and circuits.

Projects

Astromorphic Transformer Lead Student Researcher, 2022–2025. Developed a bioplausible transformer architecture leveraging neuron-astrocyte interactions to emulate self-attention mechanisms. Incorporated Hebbian and presynaptic plasticities with non-linearities and feedback, achieving superior accuracy and faster convergence on sentiment classification (IMDB), image classification (CIFAR-10), and language modeling (WikiText-2) tasks. See publication: [IEEE TCDS 2025].

Spintronic Device-Based Memory Lead Student Researcher, 2023–Present. Led end-to-end fabrication of spintronic memory arrays using e-beam lithography, sputtering deposition, and lift-off patterning. Performed comprehensive electrical characterization (I-V, R-H loops, retention, endurance) and extracted device parameters for compact model development. Advanced non-volatile memory technologies for neuromorphic computing and in-memory processing applications.

MIPS Microprocessor Design Lead Student Researcher, 2018–2019. Designed and implemented a 5-stage pipelined MIPS microprocessor in Verilog with instruction/data memory, forwarding, and hazard detection. Verified functionality through comprehensive simulation and synthesized for FPGA deployment.

Select Publications

[1] **Md Zesun Ahmed Mia**, Jiahui Duan, Kai Ni, and Abhronil Sengupta. "Trilinear Compute-in-Memory Architecture for Energy-Efficient Transformer Acceleration". In: *arXiv preprint arXiv:2604.07628* (2026).

[2] **Md Zesun Ahmed Mia**, Malyaban Bal, and Abhronil Sengupta. "Delving deeper into astromorphic transformers". In: *IEEE Transactions on Cognitive and Developmental Systems* (2025).

[3] Arnob Saha, **Md Zesun Ahmed Mia**, Jiahui Duan, Kai Ni, and Abhronil Sengupta. "Toward Variation-Tolerant Ferroelectric Neural Computing: Special Session Paper". In: *2025 IEEE 68th International Midwest Symposium on Circuits and Systems (MWSCAS)*. IEEE. 2025, pp. 583–587.

[4] Tao Zhang, Mingjie Hu, **Md Zesun Ahmed Mia**, Hao Zhang, Wei Mao, Katsuyuki Fukutani, Hiroyuki Matsuzaki, Lingzhi Wen, Cong Wang, Hongbo Zhao, et al. "Self-sensitizable neuromorphic device based on adaptive hydrogen gradient". In: *Matter* 7.5 (2024), pp. 1799–1816.

[5] KM Ashraful Hoque Fahim, Md Jubair Hasan Khalid, **Md Zesun Ahmed Mia**, and Mirza Rasheduzzaman. "Study of 3-nm Cylindrical GAAFETs with Variations in High-k Dielectric Gate-oxide Materials". In: *2022 IEEE Symposium on Industrial Electronics & Applications (ISIEA)*. IEEE. 2022, pp. 1–5.

Recognitions

- Melvin P. Bloom Memorial Outstanding Doctoral Research Award (2026)
- The Wormley Family Graduate Fellowship, Harry G. Miller Fellowships in Engineering (2025)
- Arthur Waynick Graduate Scholarship (2024),Milton and Albertha Langdon Memorial Fellowship (2023), Melvin P. Bloom Memorial Fellowship (2022)

Professional Affiliations

- Reviewer, IEEE TNNLS (2025), Design Automation Conference (DAC) (2025), IEEE MWSCAS (2025)

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- Student Member, IEEE (2015-Present), Executive Member, EDS, IEEE BD (2021-2022)